Computer Science & IT



C Programming

Language Processor

Lecture No. 01



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Topics to be Covered



Instruction Set Architecture

- Topic: Machine / Computer and ISA
- Topic: Program
- Topic: Language Processors — Compiled
- Topic: Compilers of C Language — ANSI C
- Topic: Our first C Program

Slide

Subject : programming & Data structure





Topic : Machine / Computer and ISA

t.me/AbhishekKumarpw ←

2007

2 years - HCL Technology

Abhishe sharma Jadavpur University

481.

2009 - till date

GATE

Slide

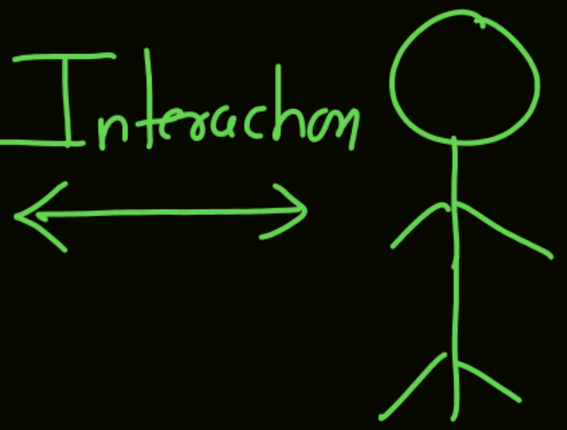
Computer — device.

processor — Instructions

Computer language — 0/1



Interaction



Communication

ISA

Instruction: ADD
SUB
MUL

Assembly language

0/1 ADD

Input → 01101100 → Computer → task complete

Assembly code

ADD : 00110100

SUB : 11011110

MUL : 11000111

Compute language 0/1 $\xrightarrow{\text{Simplify}}$ Assembly code

Single Instruction they perform only 1 operation

program Size - Big

Assembly language is Not readable / Interpretation

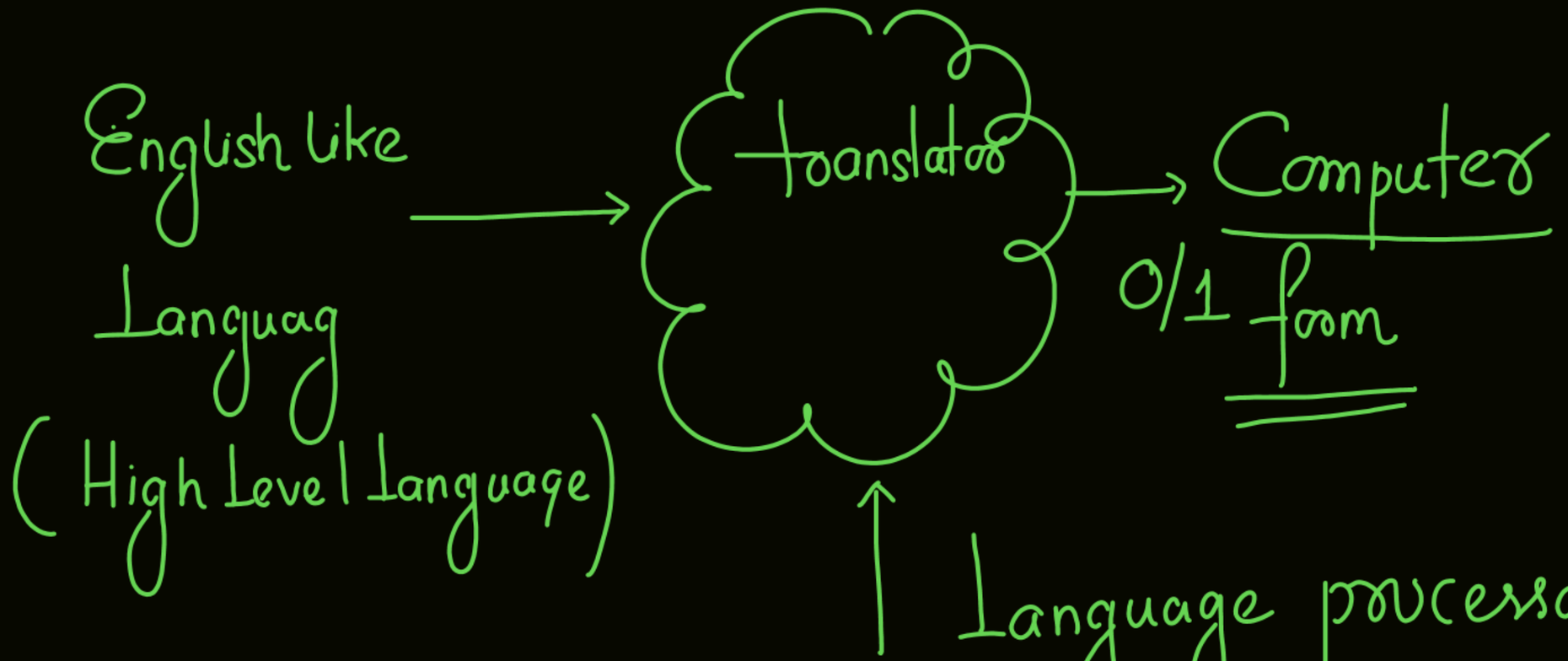
What was Required

Human understandable language : English

High level Language → Human understandable Languag. (English)

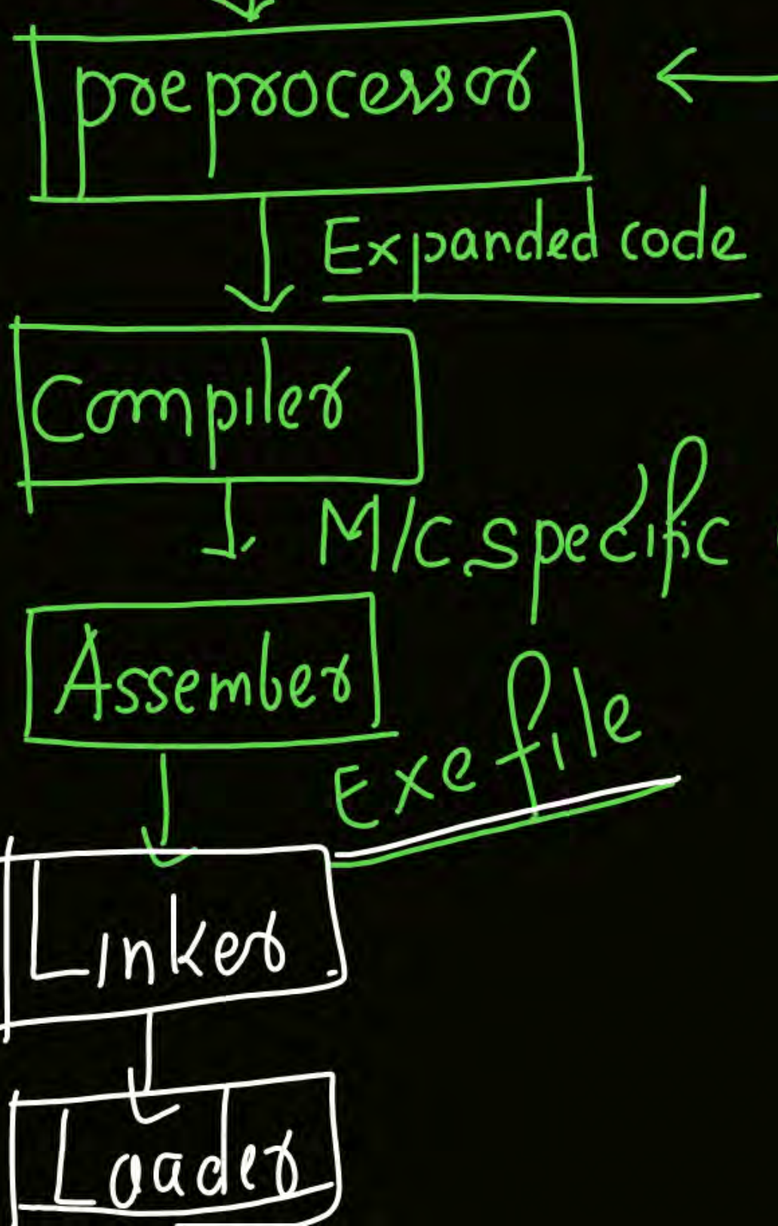
readability /

Syntax → Rules



How a High level Language translated to M/C language

program : Set of instruction (Human understandable Form)



include <stdio.h>

Input/output

macro expansion

define Max 100

max

wherever in the program Max will encounte it replace it by 100.

← Macro का expansion

↑ Symbol

#define Max 100

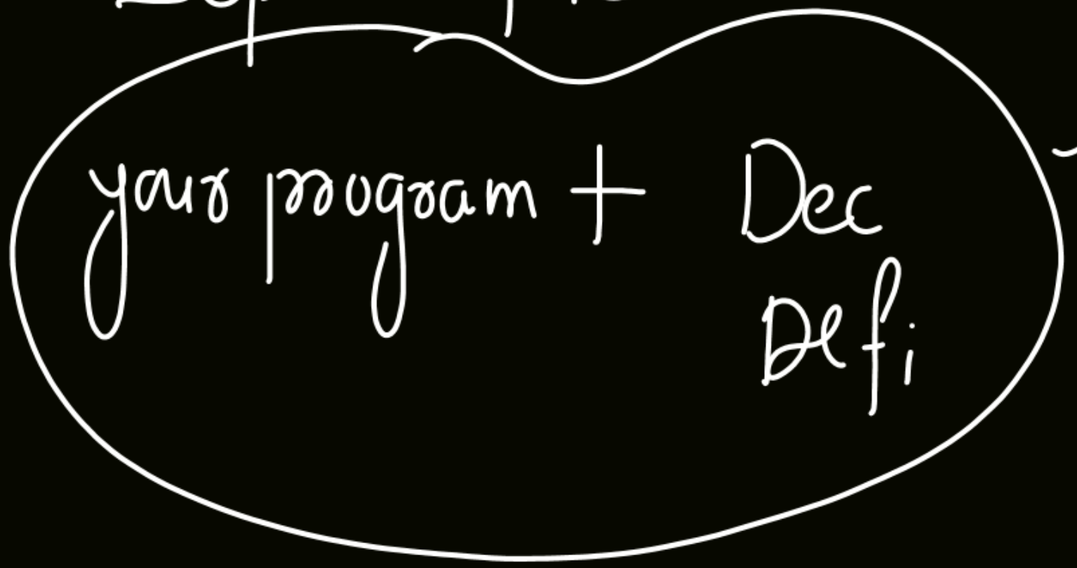
if(Max > 100)

↓ preprocessor

if(100 > 100)

Linker

The declaration & definition present in seperate file.



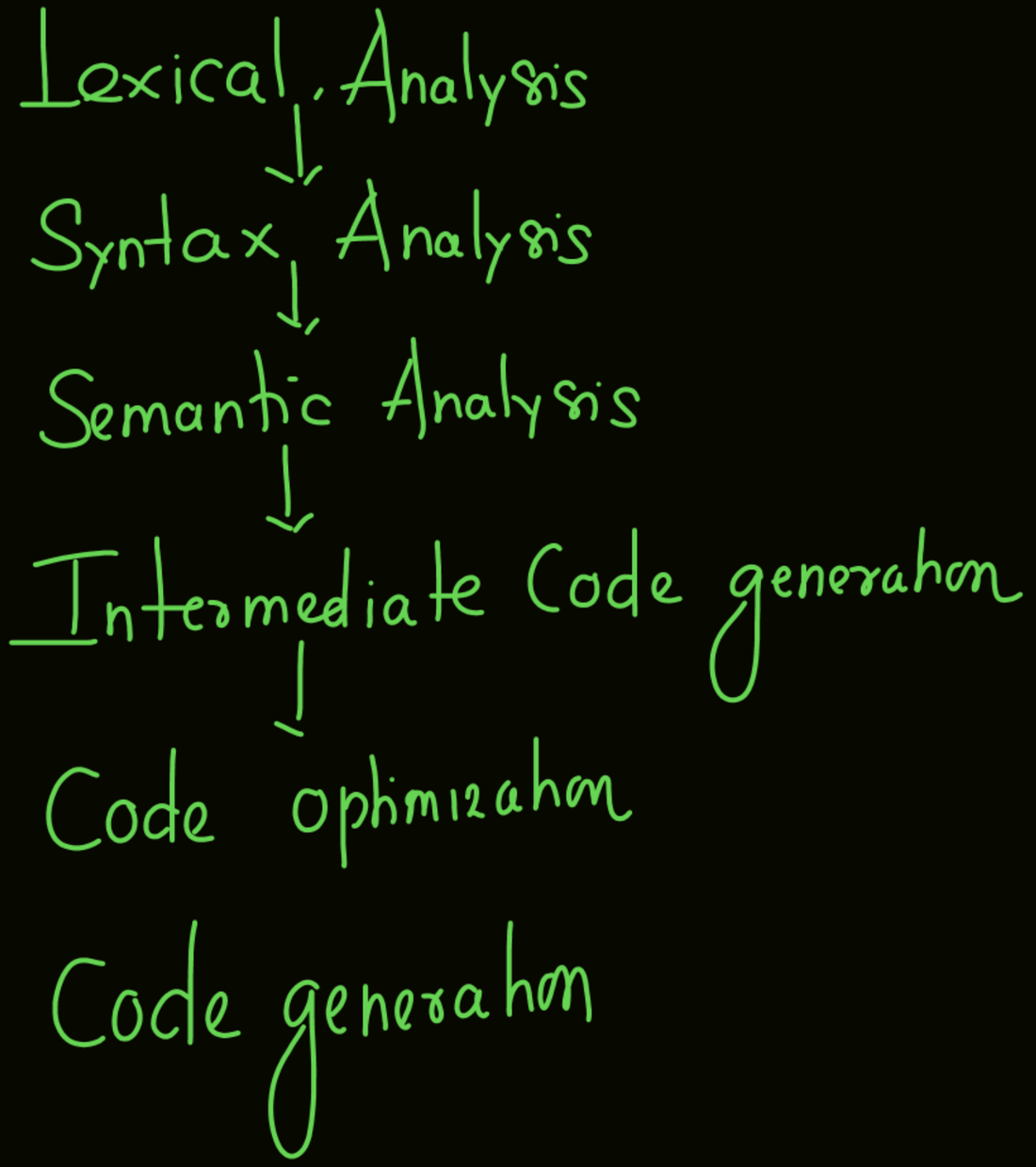

```
main() {
    :
    :
    printf(    );
}
```

No Declaration
No. Definition

Sp

Linker

Compiler :


Lexical Analysis
Syntax Analysis
Semantic Analysis
Intermediate Code generation
Code optimization
Code generation



Topic : Assembly Language

Windows program will Not Run Linux

Object code : Turbo C

Assembly language - 8085 MOV, ADD } M/C Specific

Assembler : object/Assembly

Code as input and generate exe file.

Exe file : string of 0's & 1's

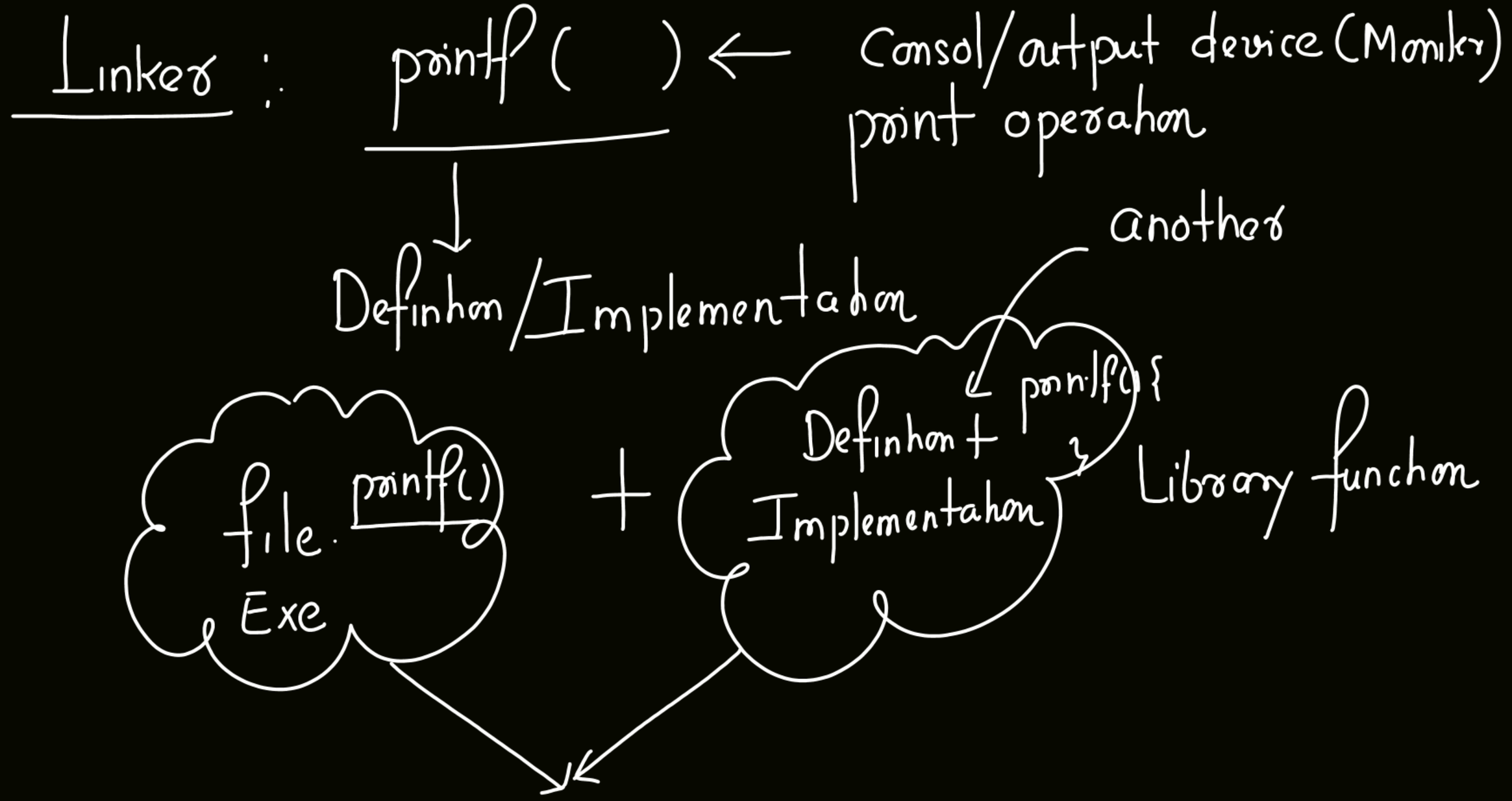
Slide

Exe ⟹ Executable file

a file which can Run on M/C consists of Instruction in form of 0/1

0101110
1100110
1110111

} Exe (Example)


Linker :
printf( )
Consol/output device (Monitor)
print operation
Definition/Implementation
another
file.
Exe
printf()
+
Definition +
Implementation
printf(){
}
Library function

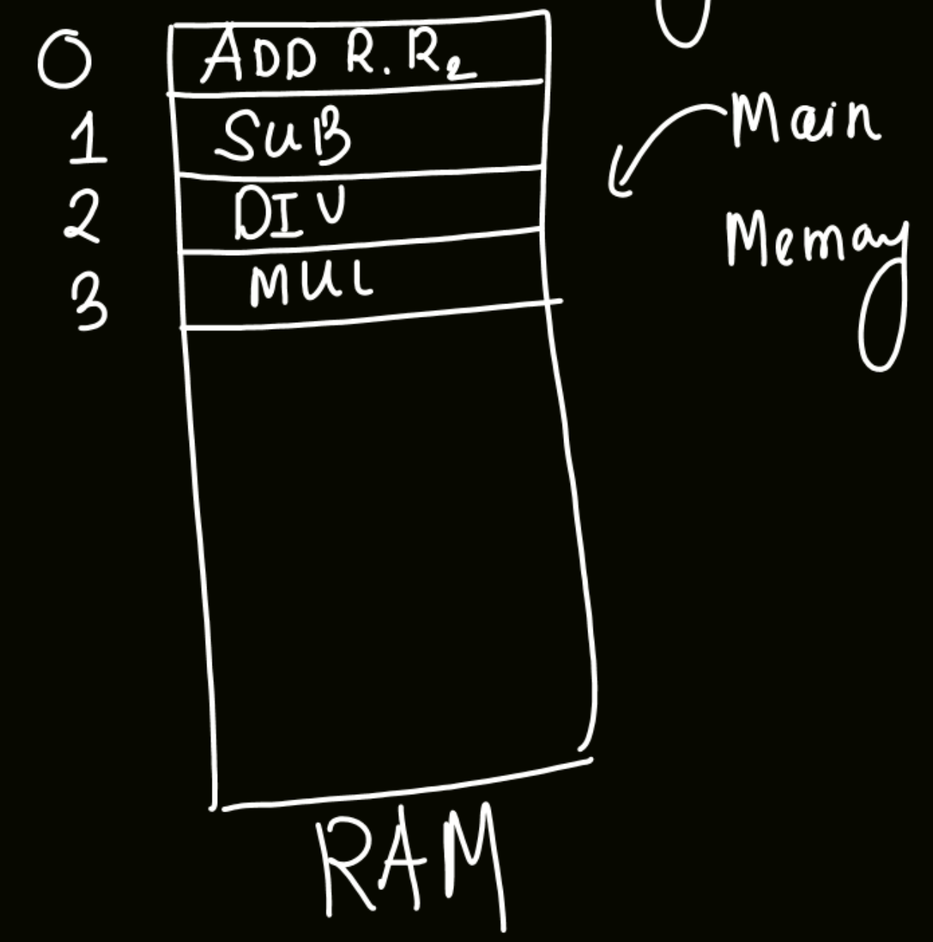
#define ← preprocessor

c

Linker responsibility is Combine all refered file together and produce Single executable file.

Loader :- program get executed in Main Memory

Single exe file (combined) ⟶ Load in memory for execution.



output of program : Sequential flow of execution

followed until some control instruction changes the flow

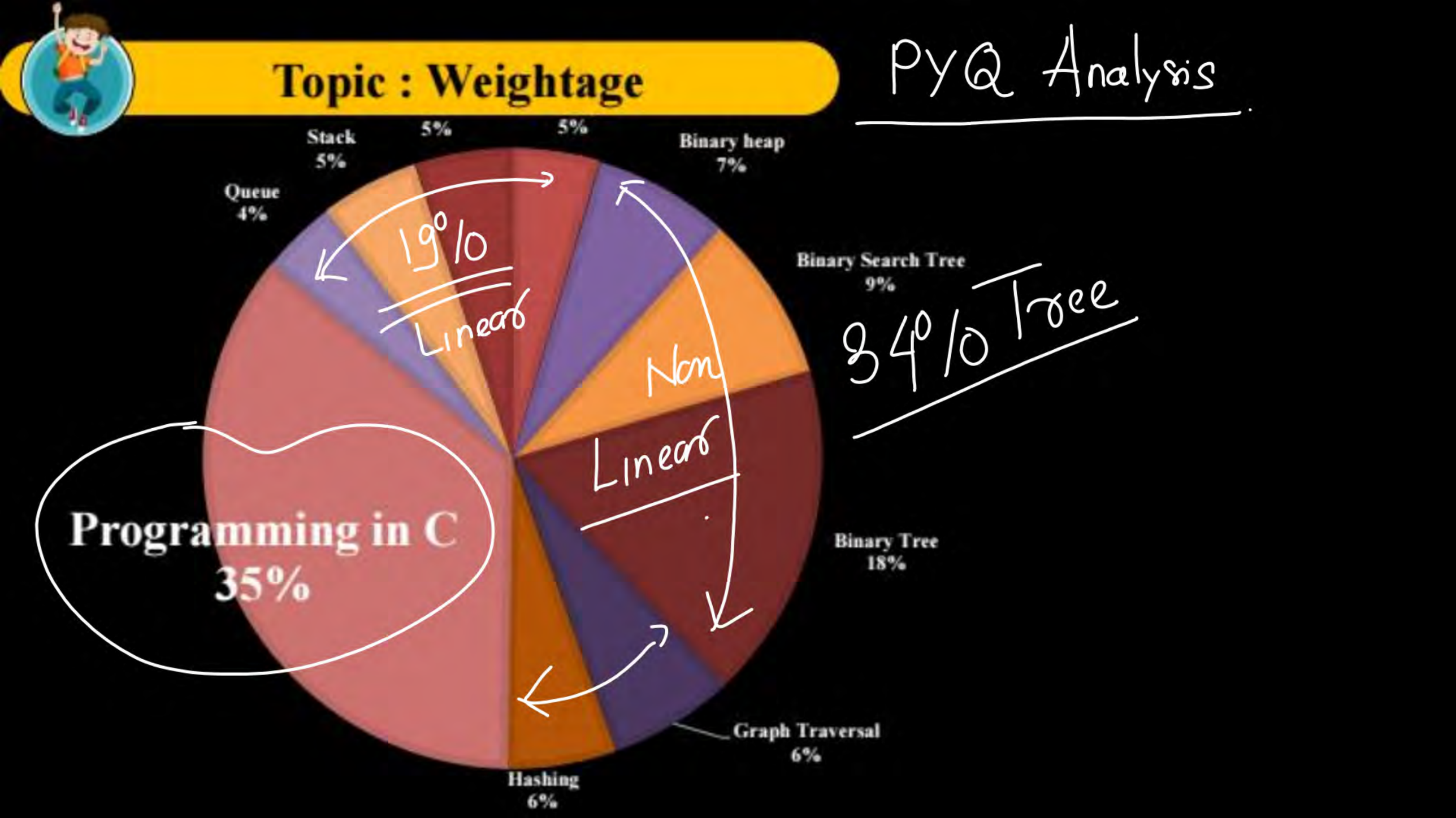
Control : jump



Topic : Syllabus

Programming in C

- Data type / Declarative statement
- operator & expression
- Input/output & expression
- Control statement
- function - Recursion
- * pointer
- * Array & pointer
- * String, Structure, union
- * Storage class
- * Dynamic allocation of memory


Topic : Weightage
PYQ Analysis
Stack
5%
5%
5%
Binary heap
7%
Queue
4%
19%
Linear
Binary Search Tree
9%
34% Tree
Non
Linear
Programming in C
35%
Binary Tree
18%
Graph Traversal
6%
Hashing
6%



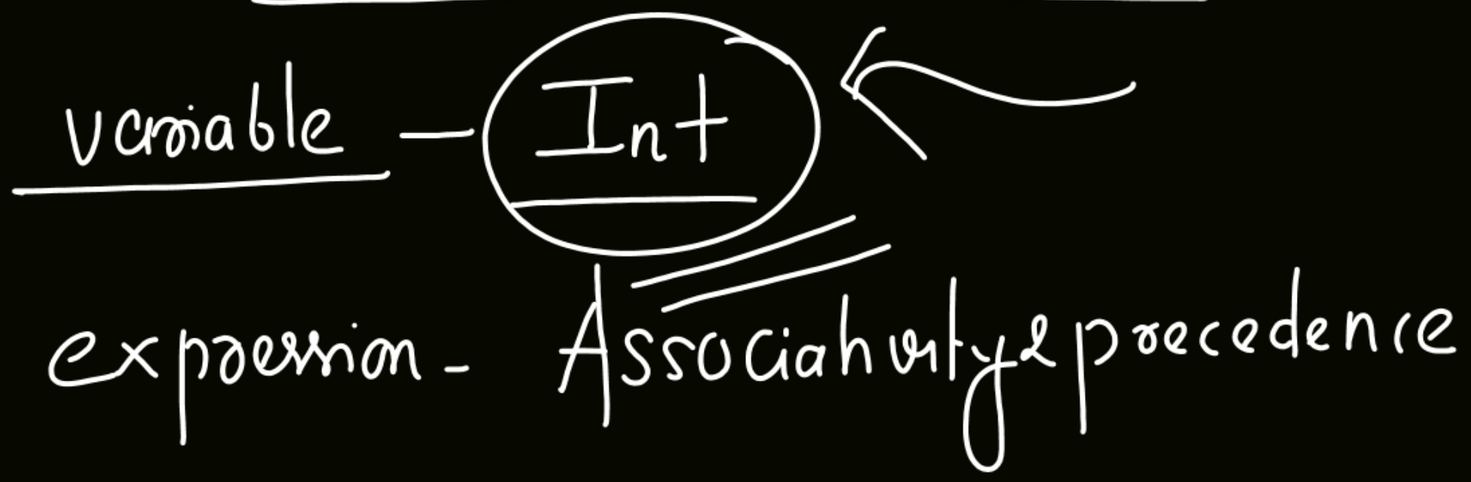
Topic : Books

Lets us `C` - poublem/challanging

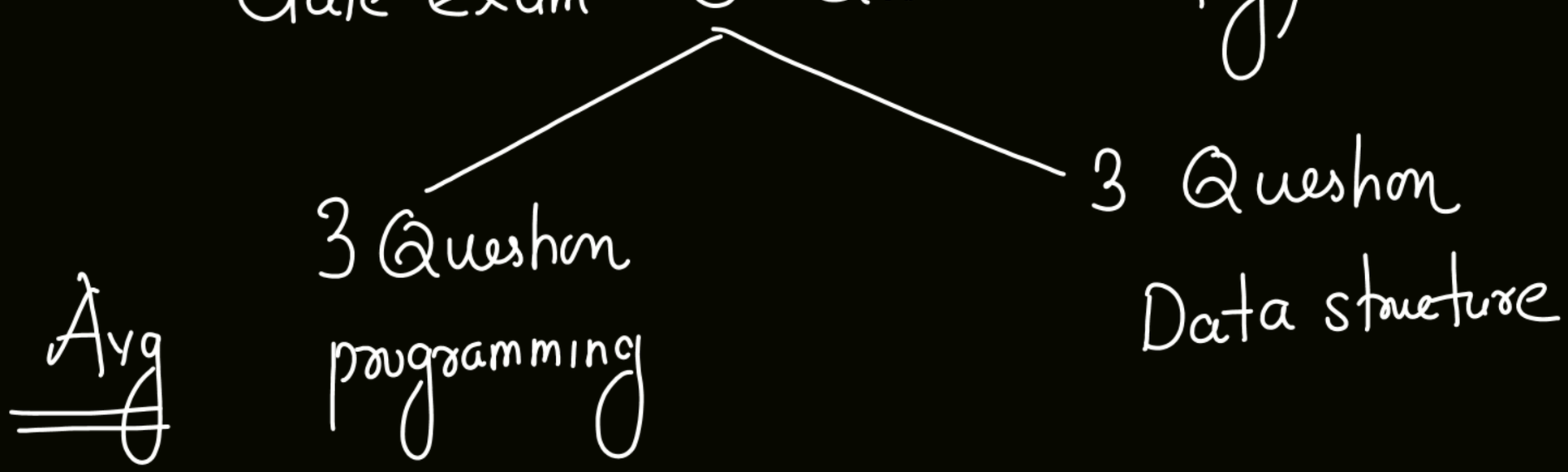
yaswant kanitkar

Dennis Ritchie

t.me/AbhishekKumarpw



Gate exam 6- Question (Avg)



Avg

PDS ⟶ Scoring — 6/6 Question correctly solve



2 mins Summary

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~~Capat~~ Capital / Capital

THANK - YOU

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